

Total No. of Questions : 8]

SEAT No. :

PC-2458

[Total No. of Pages :2

[6354]586

B.E. (Information Technology)
NATURAL LANGUAGE PROCESSING
(Elective - V) (2019 Pattern) (Semester - VIII) (414451 C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) What is Lexical Semantic? What are the main approaches to meaning representation? Compare and contrast them. [10]

b) Write a short note on: Relation of class model and state model. [8]

OR

Q2) a) What is Word Sense Disambiguation (WSD)? Explain the dictionary based approach to Word Sense Disambiguation. [6]

b) Explain in detail the types of referring expressions. [6]

c) Explain Semantic meaning and its representation with respect to Indian languages. [6]

Q3) a) Explain dependency path? How it is used in Experimental Evaluation? What is Kernel? [9]

b) What is role of annotated knowledge in text reports. [9]

OR

Q4) a) How sentences are mapped into Word Sequences and then to Dependency Paths. [9]

b) Write a note on: [9]

i) Frame semantics

ii) Semantic role labelling

P.T.O.

Q5) a) How pages are classified for trainable document separation. **[9]**

b) Write a note on : **[9]**

- i) Domain Knowledge
- ii) Domain Concepts
- iii) Knowledge Roles

OR

Q6) a) Write in detail the high level representation approaches in text mining. **[9]**

b) What is probabilistic model? Give examples of probabilistic models and explain any one in detail. **[9]**

Q7) a) Write short note on :

- i) Stemmers
- ii) Part-of-Speech Tagger. **[10]**

b) Explain design feature of IR with a neat diagram. **[6]**

Q8) a) With a suitable example explain cluster based IR modeling. **[9]**

b) Explain in detail "Evaluation of the IR System". **[7]**

